SUJAN CHAPAGAIN

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EDUCATION

BACHELOR OF SCIENCE IN AGRICULTURE 2021-2025
Tribhuvan University, Institute of Agriculture and Animal Science, Lamjung Campus
CGPA: 83.86% ● Cohort topper

PUBLICATIONS

1. **Chapagain, S**., Shrestha, S., Kandel, B. P., and Rana, D. (2026). Phenotypic screening for leaf rust resistance in wheat genotypes under open-field conditions. (*Under 2nd revision*).
2. **Chapagain, S**., Kandel, B.P., Shrestha, S., et al. (2026). Screening drought tolerance capacity in wheat genotypes using physiological traits and drought tolerance indices in the mid-hills of Nepal. Advances in Agriculture. https://doi.org/10.1155/aia/4020245
3. **Chapagain, S**., Shrestha, S., Kandel, B. P., Rana, D., & Poudel, A. (2026). Characterization of phenotypic diversity and trait associations in Nepalese finger millet landraces. *CABI Agriculture and Bioscience*. https://doi.org/10.1079/ab.2026.0044
4. **Chapagain, S**., Kandel, B.P., Shrestha, S. et al. (2026). Deciphering genetic variability and trait associations in Nepalese finger millet landraces for yield improvement. Discover Agriculture 4, 130. DOI: 10.1007/s44279-026-00587-3
5. **Chapagain, S**., Kandel, B. P., Poudel, A., et al. (2026). Assessment of early seedling stage indices for the selection of potential drought-tolerant bread wheat genotypes. Discover Applied Science. DOI: 10.1007/s42452-026-08252-6
6. **Chapagain, S**., Kandel, B. P., Adhikari, A., Poudel, A., & Kandel, M. (2025). Estimation of genetic parameters and trait association in wheat (*Triticum aestivum* L.) genotypes. Journal of the Institute of Agriculture and Animal Science, 39(1), 151–155. DOI: 10.3126/jiaas.v39i1.88144
7. **Chapagain, S.**, Kandel, B. P., Shrestha, S., Poudel, A., & Yadav, S. P. S. (2025). Identifying superior finger millet (*Eleusine coracana* L.) landraces using the multi-trait genotype-ideotype distance index (MGIDI). Cogent Food & Agriculture, 11(1). DOI: 10.1080/23311932.2025.2592358

PRESENTATIONS

1. Chapagain, S., Kandel, B. P., Poudel, A., & Subedi, M. (2025, January 13). Morphophysiological screening of wheat genotypes under water deficit conditions. 1st National Convention of Plant Breeding, Genetics and Seed Science, Lalitpur, Nepal. https://doi.org/10.13140/RG.2.2.28442.50885
2. Chapagain, S., Kandel, B. P., Poudel, A., & Subedi, M. (2025, November). Phenotypic screening for leaf rust resistance in wheat genotypes under open-field conditions. 13th Research Practice and Seminar Symposium (RPS), Lamjung Campus, Nepal.

GRANT

Undergraduate Agriculture Research Mini Grant 2025 – 2026
Physiological response and agronomic performance of wheat genotypes under drought stress
Funding agency: Champadevi Rural Municipality | Funding amount: USD 2200

SKILLS

- Field and greenhouse skills: Trial layout and experimental design (CRD, RCBD, Split-plot, and Alpha-lattice); pathogen inoculation; controlled stress screening; phenotyping; hybridization and cross-pollination
- Physiology: chlorophyll fluorescence, relative water content, malondialdehyde
- Molecular and laboratory techniques: DNA/RNA extraction and quantification; PCR; gel electrophoresis; co-expression analysis, NCBI BLAST, and Ensembl Plants
- Genetic diversity analyses: Population structure and genetic diversity analyses (PCA, DAPC, AMOVA); G×E interaction analysis
- Bioinformatics and statistical tools: R, Python, Linux, GenStat

EXPERIENCES

Nepal Agricultural Research Council, Khumaltar, Lalitpur 2025 Nov - Ongoing
Supervisor: Mahesh Subedi (Scientist S2), Dryland Lab

- Evaluated 200 wheat genotypes under a single-row design for leaf rust resistance screening.
- Extracted genomic DNA from wild rice accessions (CTAB method) and conducted PCR amplification and gel electrophoresis for molecular marker analysis.
- Phenotyped agronomic (plant height, tiller number, spike length) and physiological traits (disease severity, vigor)

Impact of cover crops on plant yield, insect pests, weeds, and soil health in organic vegetable production

Volunteer (Assistant) 2024-2025
Funded by: USDA-FAS-Tennessee State University & IAAS, Tribhuvan University

- Collected field data on crop performance, management practices, and farmer observations.
- Prepared field reports and summarized results for project documentation and evaluation.

MEMBERSHIPS

American Phytopathological Society (APS) 2026, Mar 5
Plant Breeding and Genetic Society, Nepal 2025, Dec 10

HONORS AND AWARDS

Best Poster Presenter, 1st National Convention of Plant Breeding, Genetics and Seed Science Jan 2026
Best Oral Presenter, 13th Research Practice and Seminar Symposium (RPS) Nov 2025
University Full Merit Scholarship, Tribhuvan University 2021–2025
Golden Jubilee Scholarship Scheme (GJSS), Embassy of India 2021–2023

VOLUNTEER AND LEADERSHIP

Director, International Association of Students in Agricultural and Related Sciences (IAAS) 2023–2024
Plant Science Instructor, Smart Education Academy 2022–2023